

Answer Key — Exploring the Investigative World of Science

Final answers with brief reasons.

Objective

- Q1** (b) **Does adding more yeast make dough rise faster?** — it can be tested by changing yeast amount and measuring rise.
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- Q2** (b) **quantitative observation** — it has a number and a unit.
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- Q3** (c) **changed (independent) variable** — the one you set yourself.
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- Q4** (a) **True** — only one variable changes; the rest stay the same.
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- Q5** (a) **Both true, R explains A** — "because it had no light" is reasoning — an inference, as R says.
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- Q6** **measured** — the dependent/measured variable.
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- Q7** **systematic** — the careful step-by-step method.
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- Q8** (a) **to check the result is reliable** — agreeing repeats show the result is reliable.
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Short answer

- Q9** Observation = directly noticed ("the bread is mouldy"); inference = worked-out explanation ("it spoiled because it was kept warm").
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- Q10** Changed = amount of fertiliser; Measured = plant height.
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- Q11** Change one variable at a time; keep careful notes; repeat to confirm (any two).
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Long / application

- Q12** Changed = water temperature; Measured = time to dissolve; Controlled = amount of sugar, amount of water, stirring. Use same sugar/water, stir both equally, change only temperature, repeat.
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- Q13** Observations: "it doubled in size," "it smelled sour." Inference: "because tiny organisms acted on it." New question e.g. "Does it rise faster in a warmer place?"
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